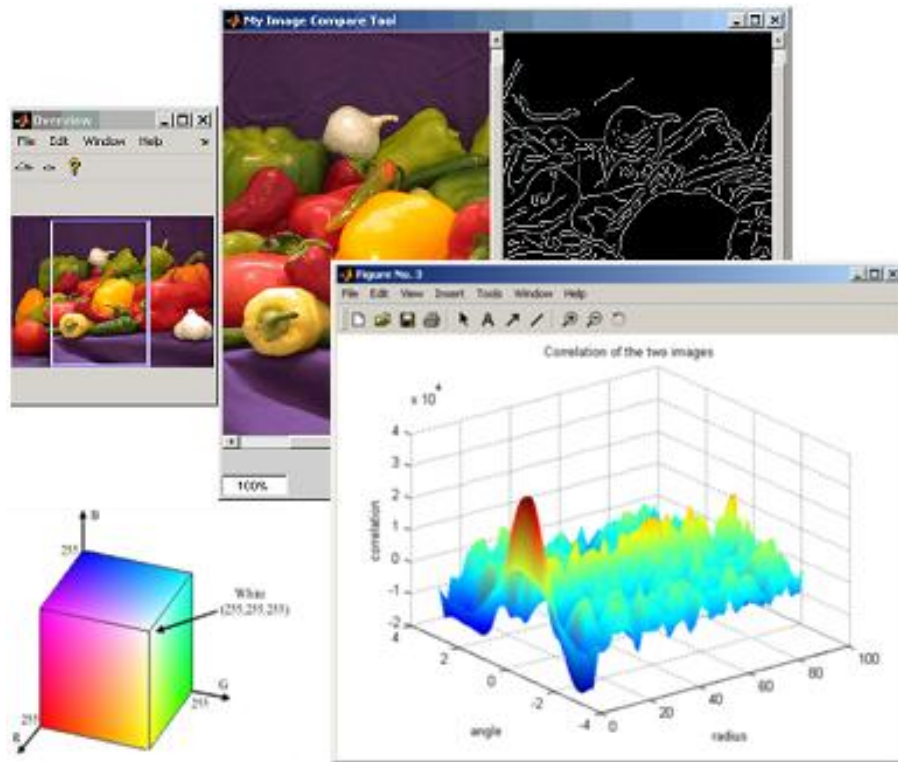


Digital Image Processing

LECTURE 15: MORPHOLOGICAL OPERATION (PART3)



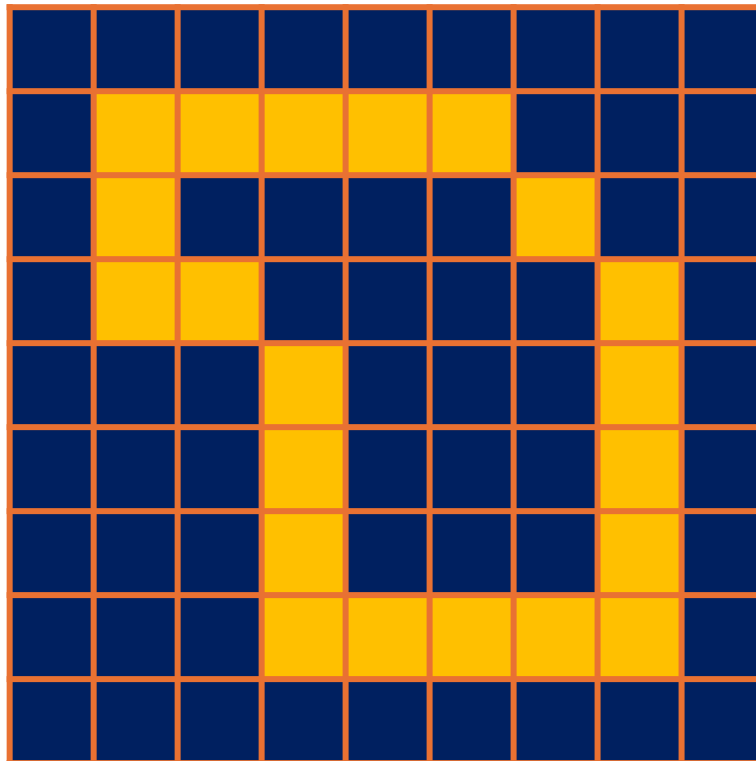
Dr. Ajay Kumar Mahato
Assistant Professor
ECE Department
GLA University Mathura

Region Filling

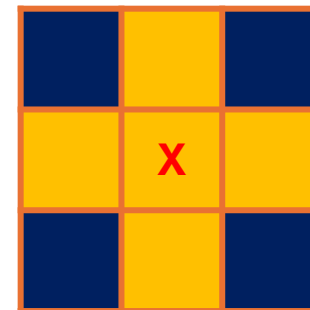
Region Filling

Example 1:

A



B



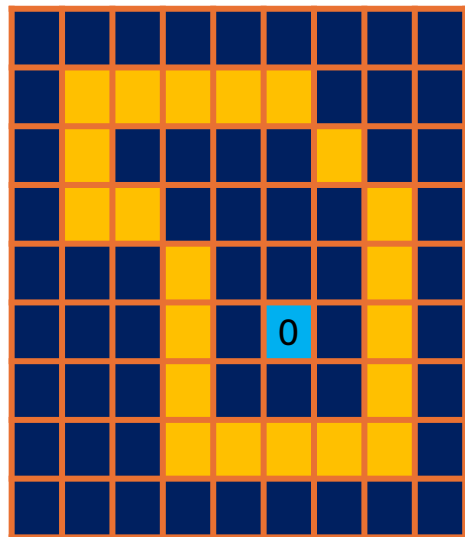
Region Filling

$$X_K = (X_{K-1} \oplus B) \cap A^c$$

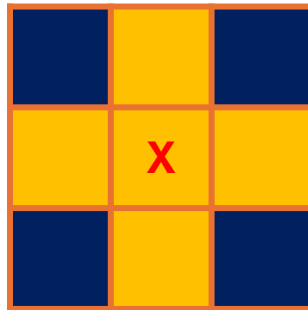
For K=1

$$X_1 = (X_0 \oplus B) \cap A^c$$

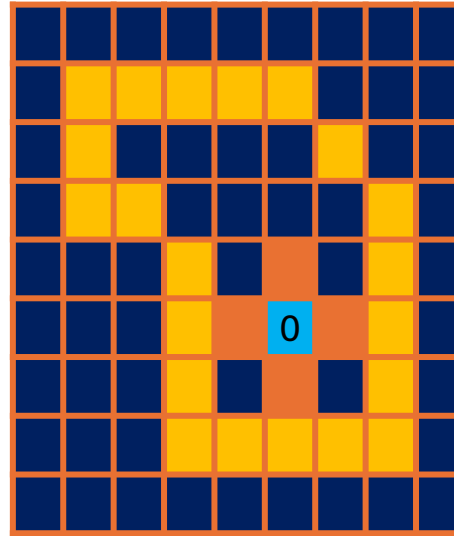
X_0



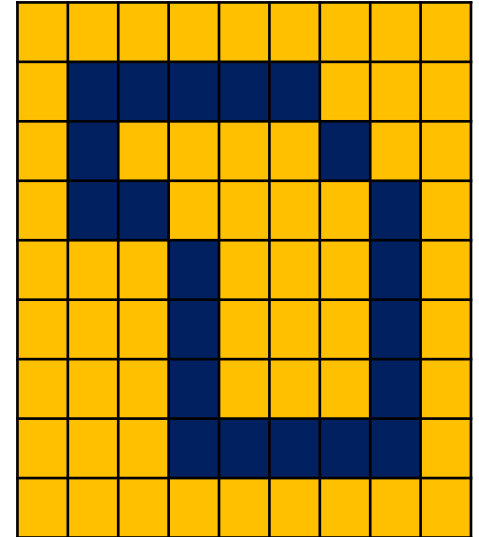
B



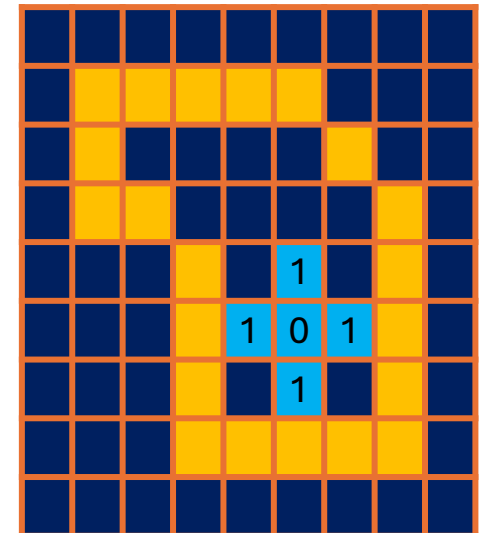
$(X_0 \oplus B)$



A^c

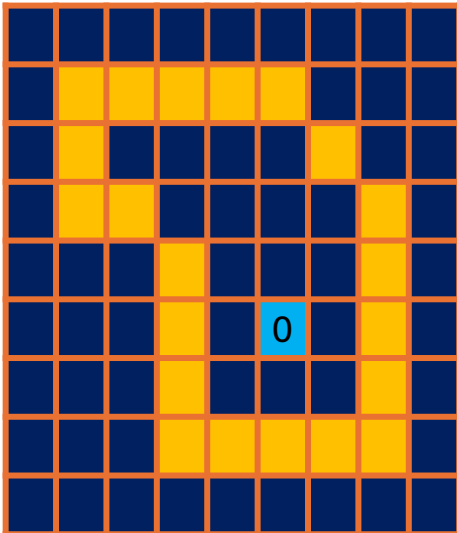


$$X_1 = (X_0 \oplus B) \cap A^c$$

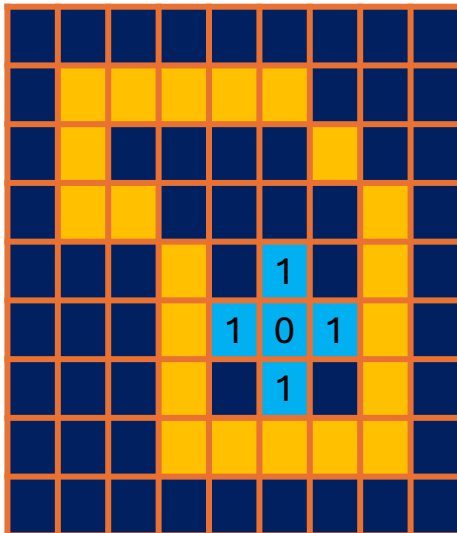


Region Filling

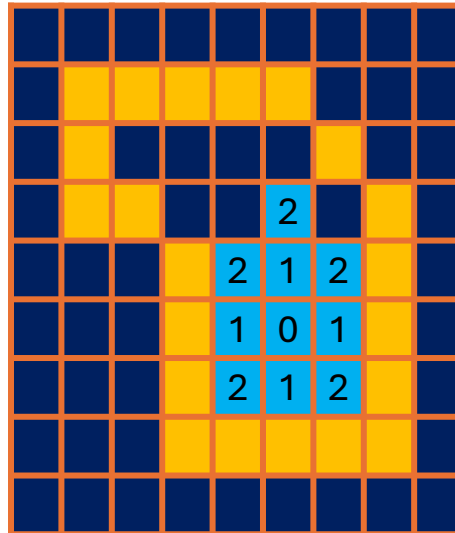
X_0



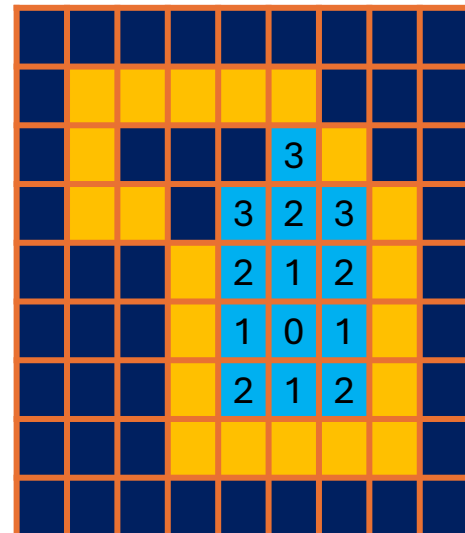
X_1



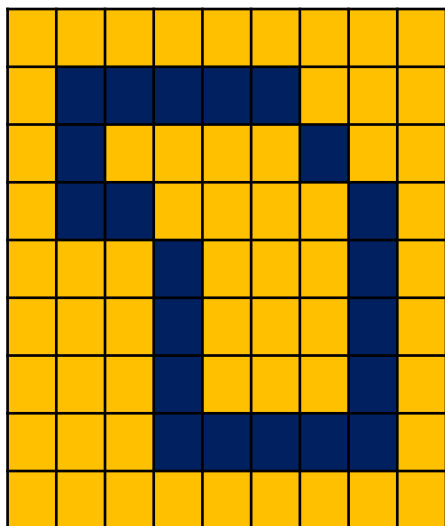
X_2



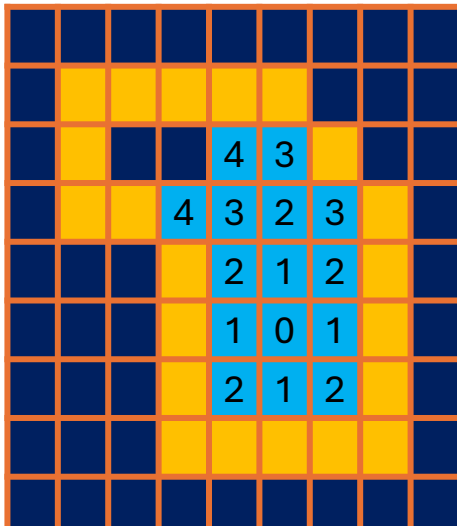
X_3



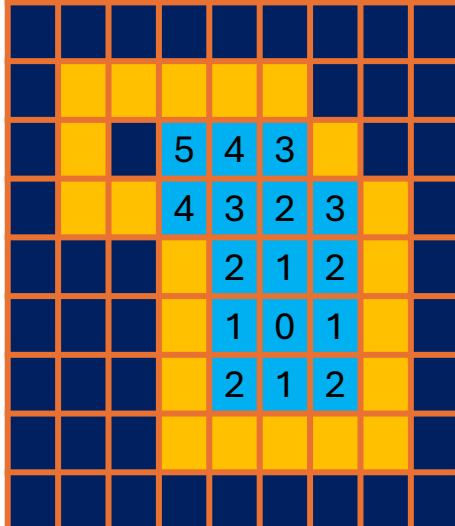
A^c



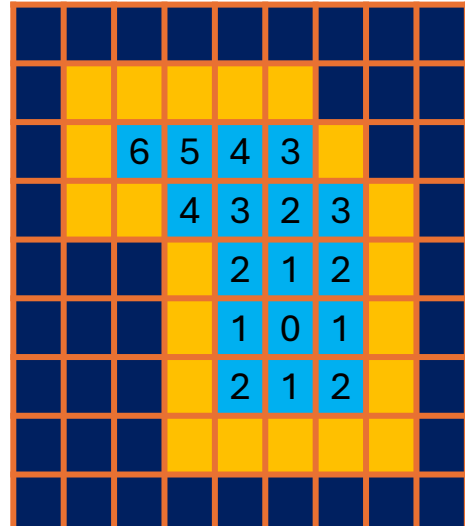
X_4



X_5



X_6



Extraction of Connected Components

Extraction of Connected Components

❑ Extraction of Connected Components

- Connected component labeling is used in computer vision to detect connected regions in the images
- It groups the pixels into components based on the pixel connectivity
- This is similar to the region filling operation. Let us assume that p is a point of the connected component C .
- The process is the same as region filling except that the original image rather than the complement is used.

Extraction of Connected Components

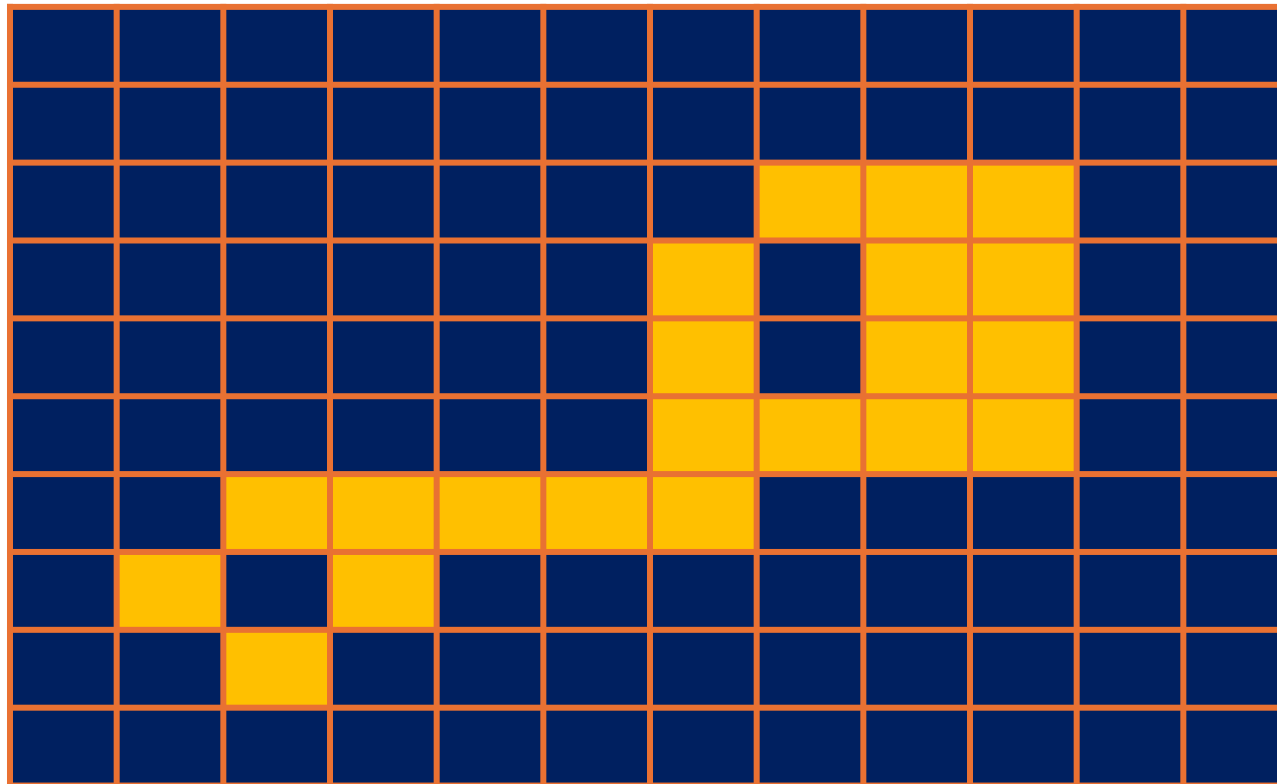
□ The process that achieve this is as follows:

1. Let P be a point in connected component C that needs to be extracted. Let $P = X_0$.
2. Let B be the structuring element.
3. Repeat steps 4 to 6.
4. $K = K+1$
5. $X_K = (X_{K-1} \oplus B) \cap A$
6. If $X_K = X_{K-1}$ then stop
7. Exit

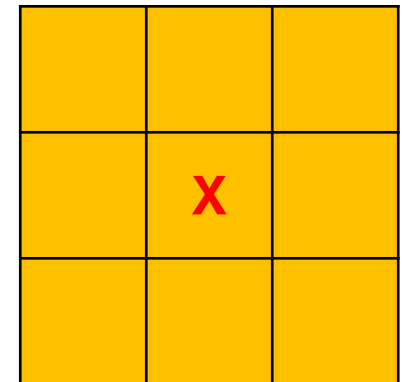
Extraction of Connected Components

Example 1:

A

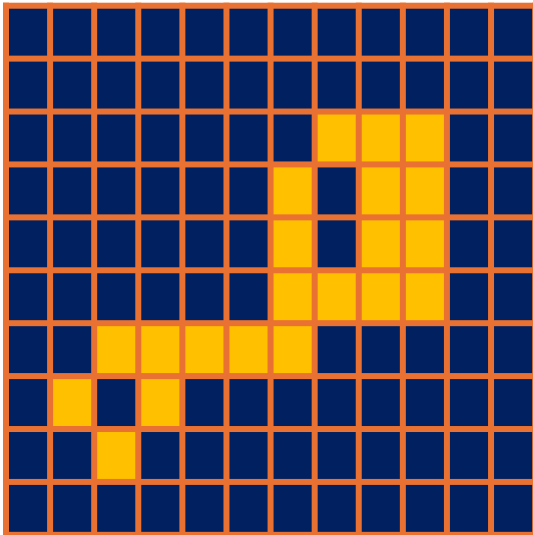


B

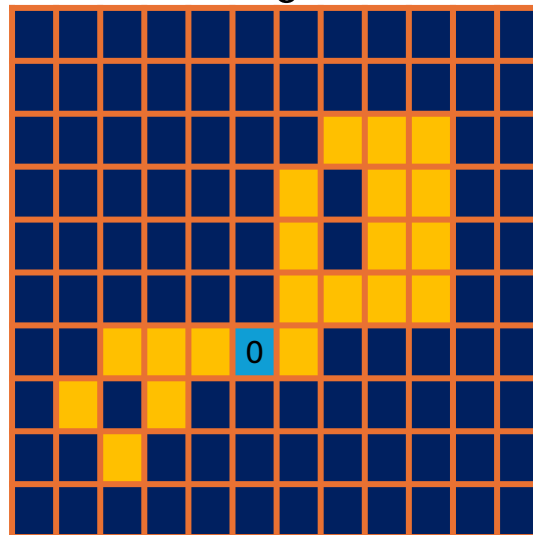


Extraction of Connected Components

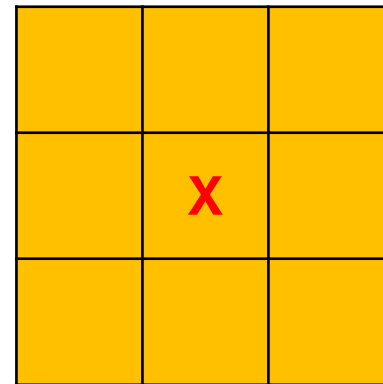
A



X_0



B

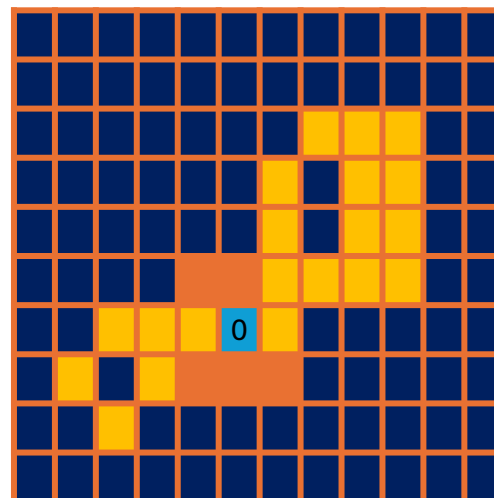


$$X_K = (X_{K-1} \oplus B) \cap A$$

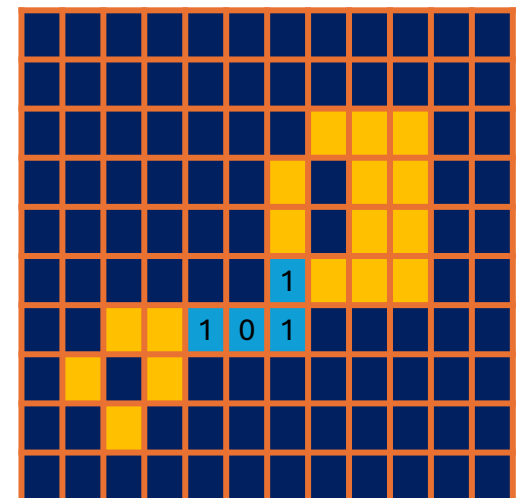
For $K=1$,

$$X_1 = (X_0 \oplus B) \cap A$$

$(X_0 \oplus B)$

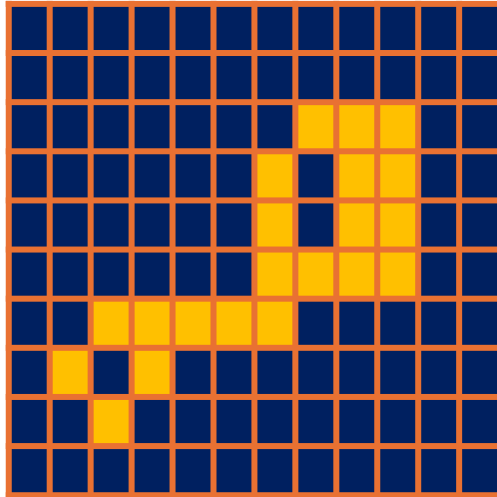


X_1

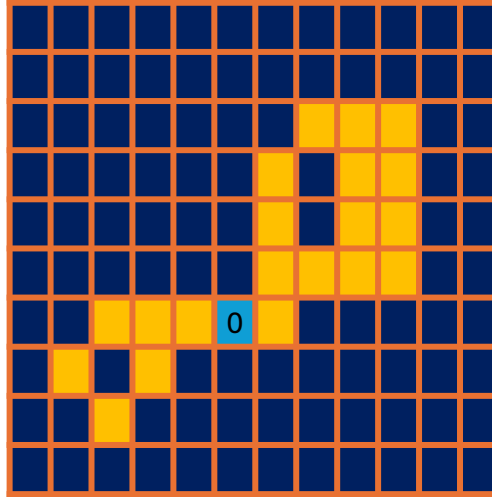


Extraction of Connected Components

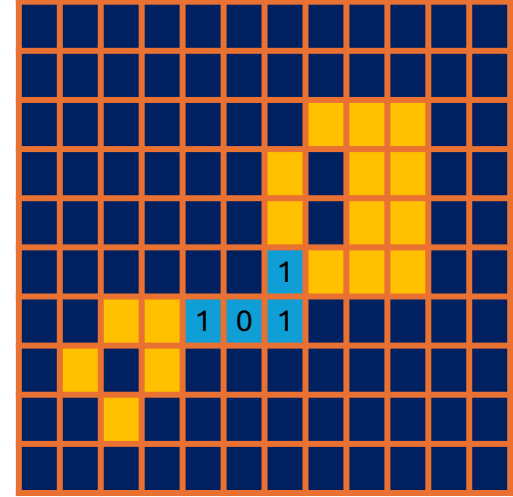
A



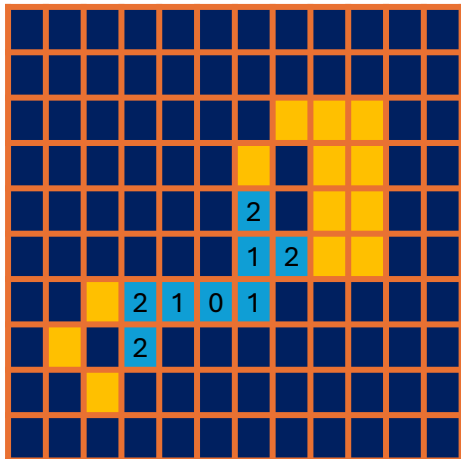
X_0



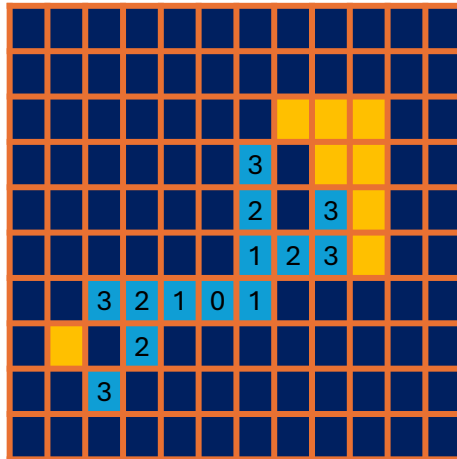
X_1



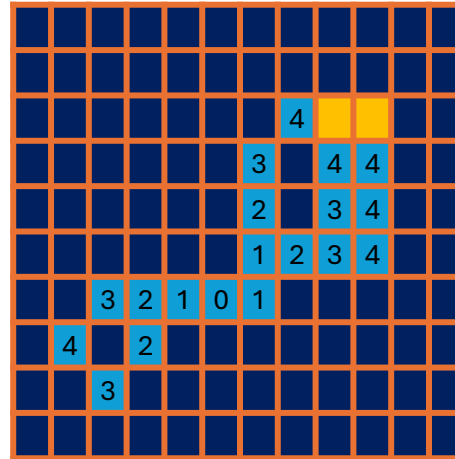
X_2



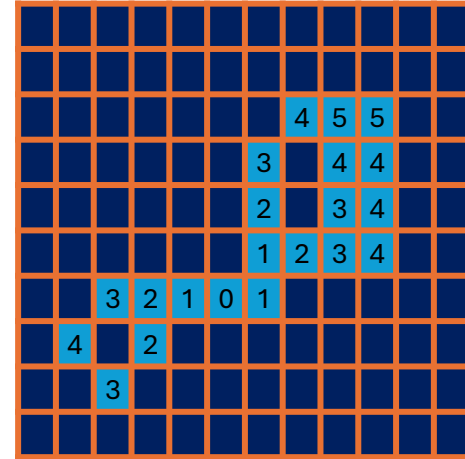
X_3



X_4

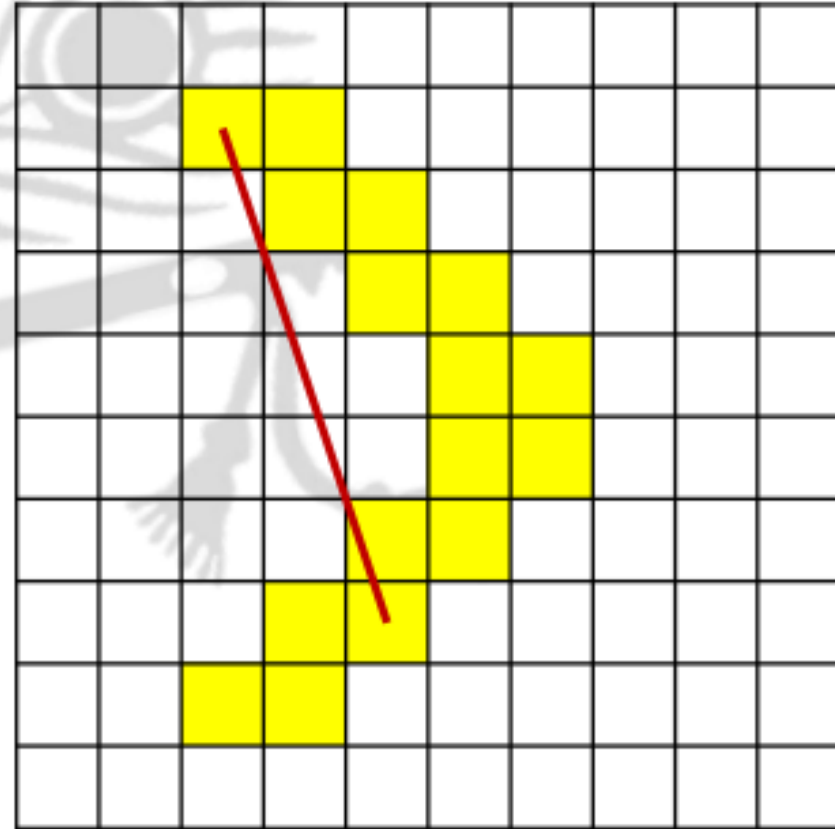
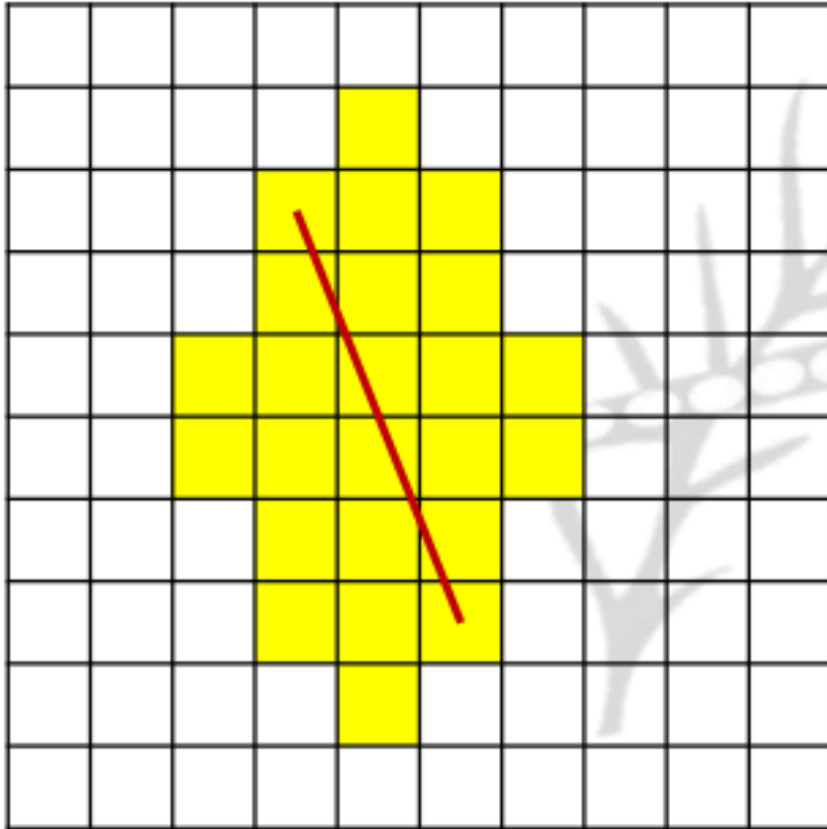


X_5



Convex Hull

Convex Hull



Convex Hull

□ The procedure consists of implementing the following equation:

- $X_0^i = A$

- $X_{k+1}^i = (X_k^i \circledast B^i) \cup A \quad i = 1, 2, 3, 4 \quad \text{and} \quad k = 1, 2, 3, \dots$

- $C(A) = \bigcup_{i=1}^4 D^i, \quad D^i = X_{converged}^i$

- Where A represents iteratively applying the hit-or-miss transform with the 4 masks.
- When the masks produce no more changes, the union is performed between all 4 masks and the result becomes D^i

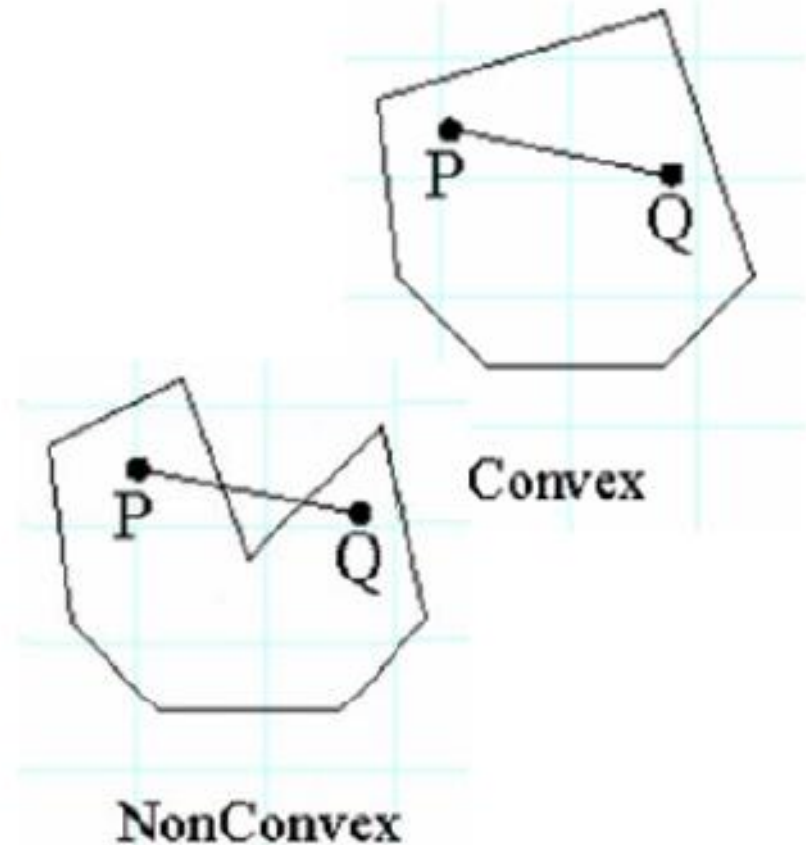
Note: converged means when

$$X_k^i = X_{k+1}^i$$

Convex Hull

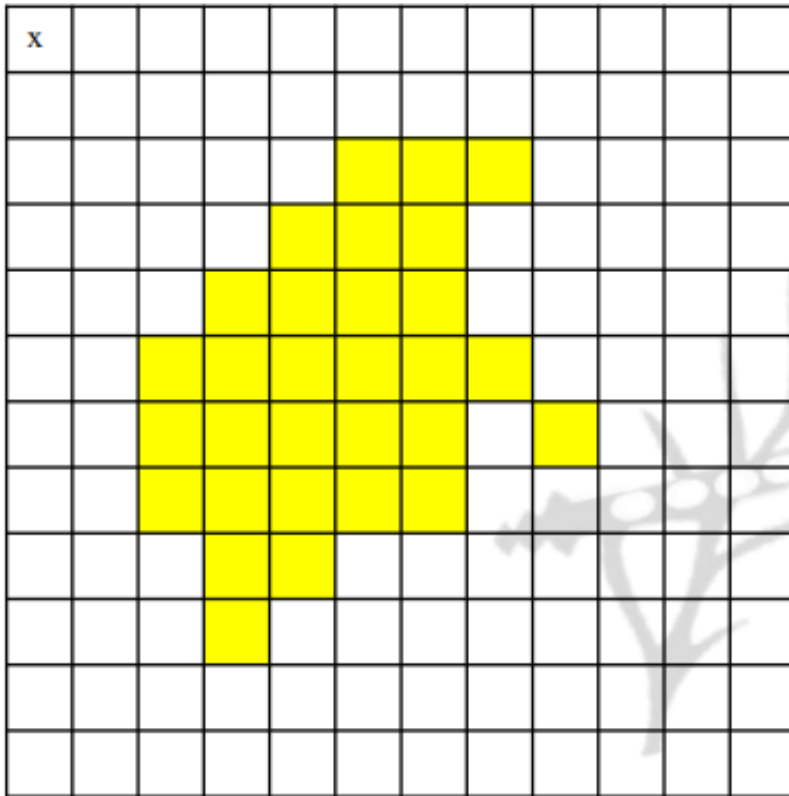
Set A is said to be **convex**, if the straight line segment joining any two points in A lies entirely within A

- The **convex hull** H of an arbitrary set S is the smallest convex set containing S
- The set difference $H - S$ is called the **convex deficiency** of S
- **Convex hull & Convex deficiency**
useful for object description



Convex Hull

Example 1:



$\leftarrow I$

$S \rightarrow$

1	x	x
1	0	x
1	x	x

S_0

1	1	1
x	0	x
x	x	x

S_1

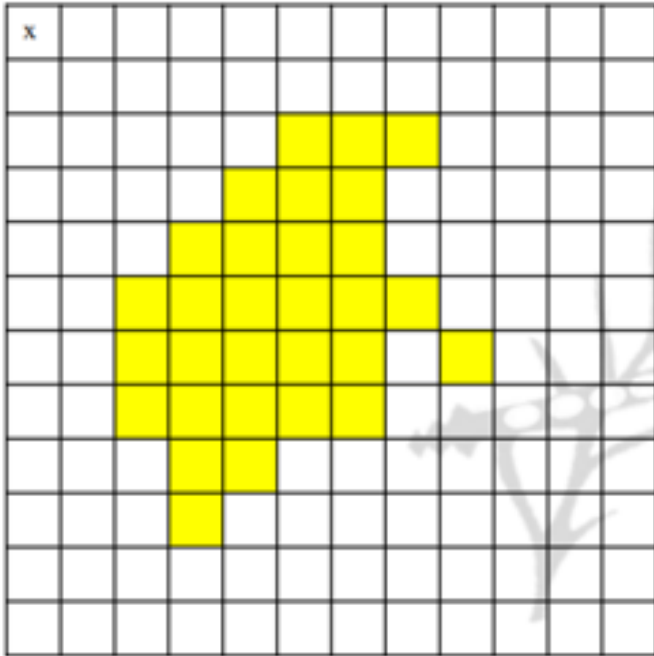
x	x	1
x	0	1
x	x	1

S_2

x	x	x
x	0	x
1	1	1

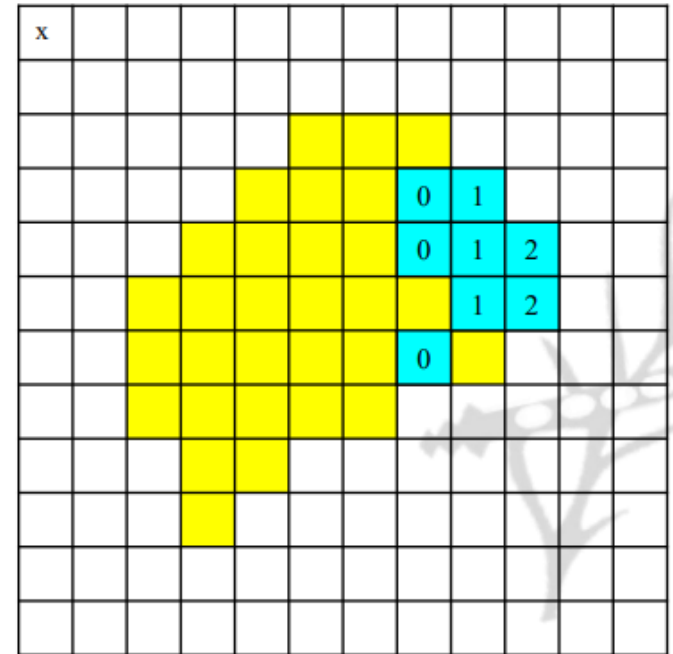
S_3

Convex Hull

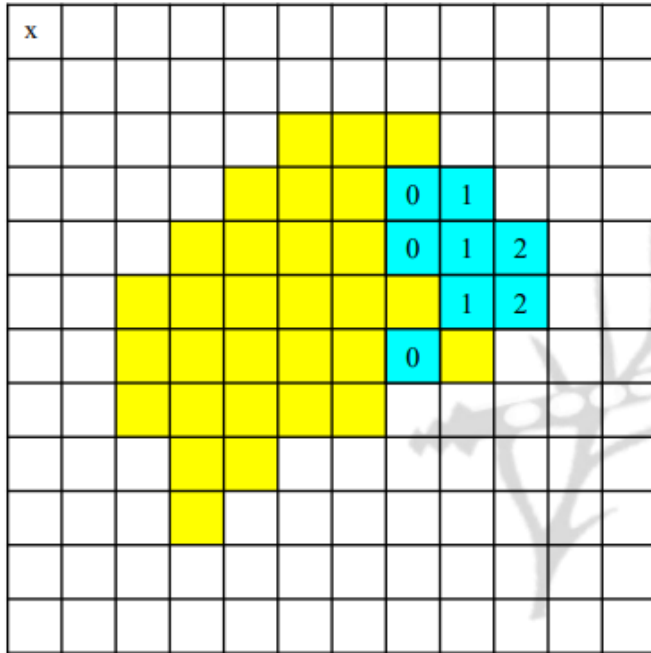


1	x	x
1	0	x
1	x	x

S_0

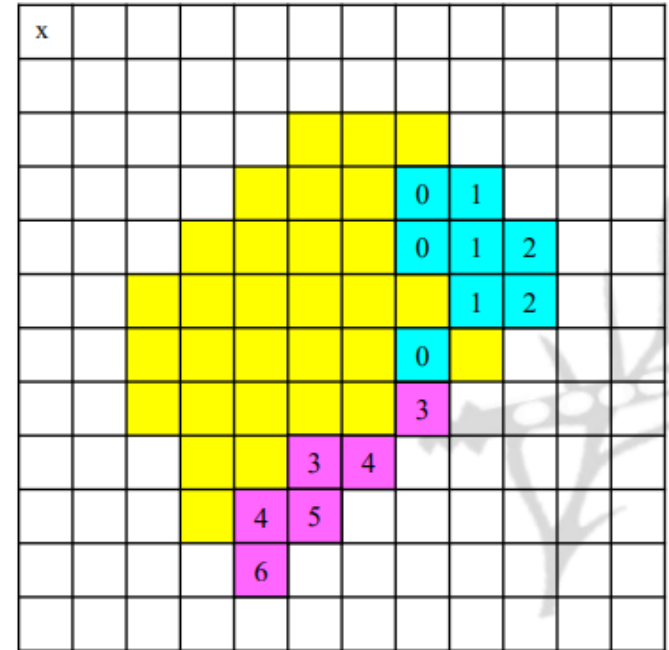


Convex Hull

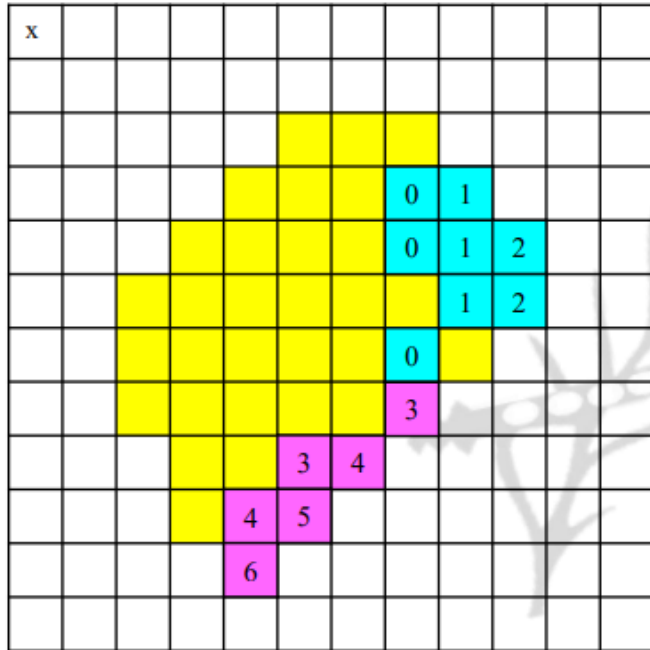


1	1	1
x	0	x
x	x	x

S_1

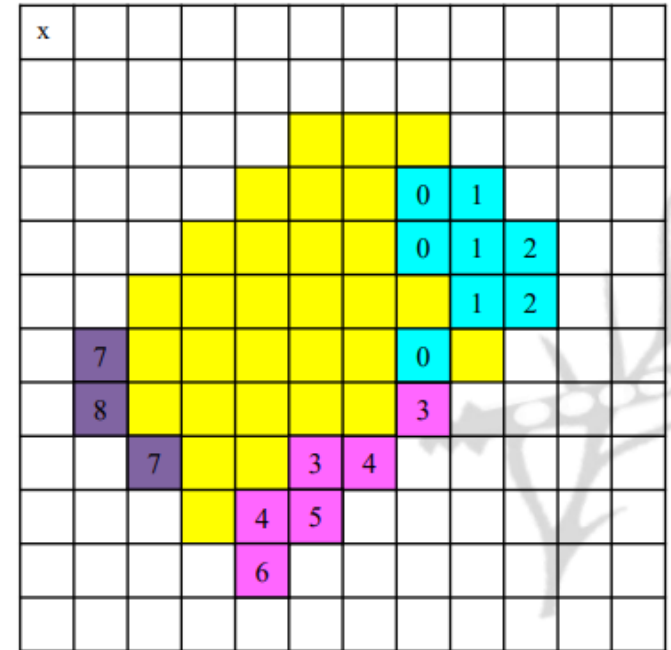


Convex Hull

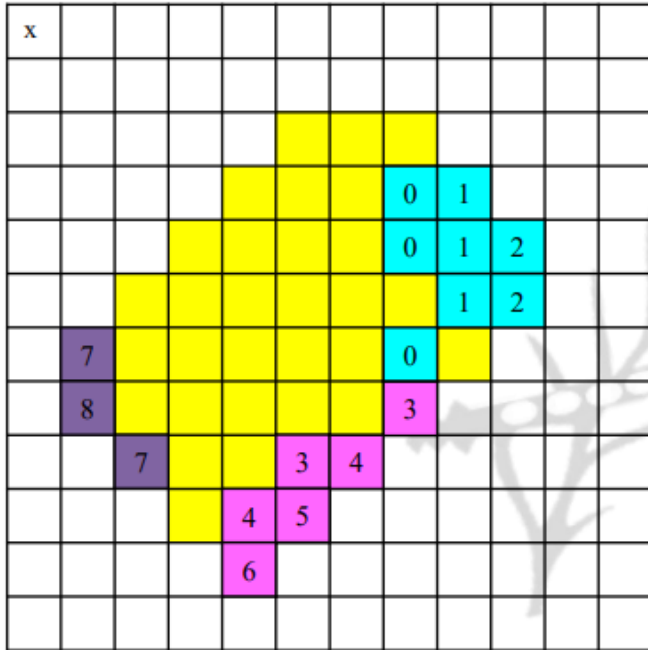


x	x	1
x	0	1
x	x	1

S_2

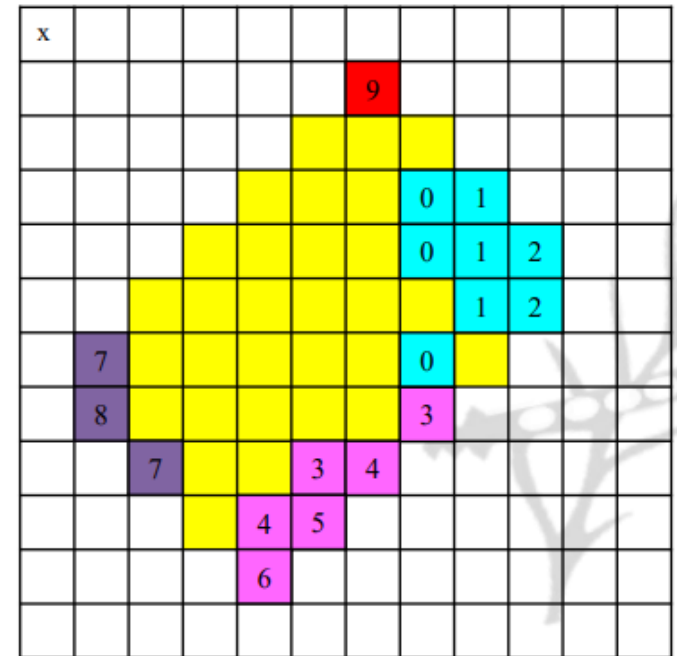


Convex Hull

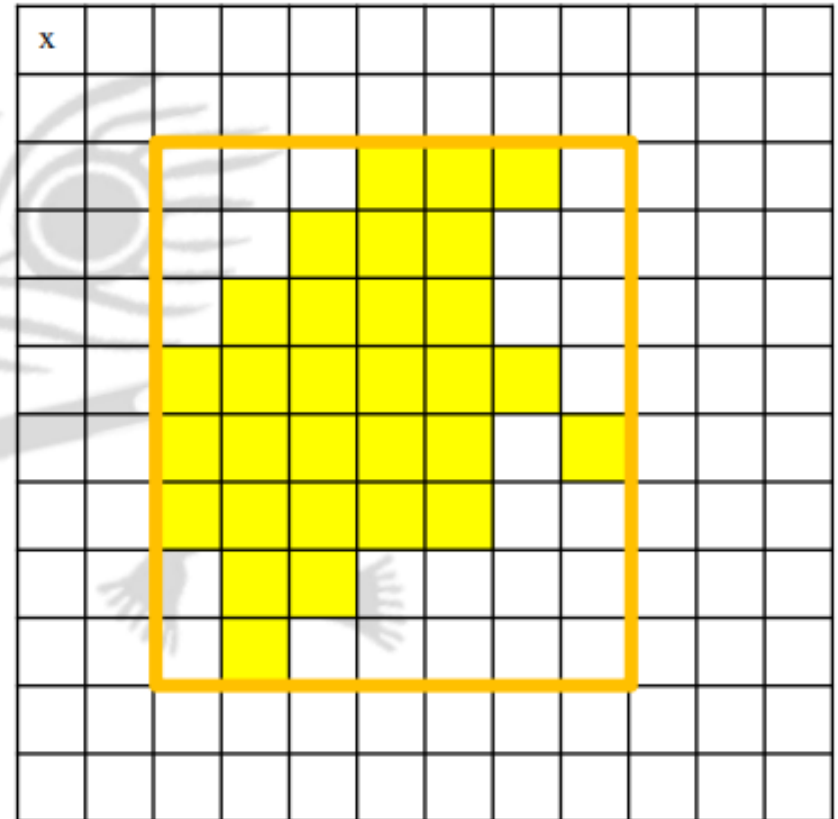
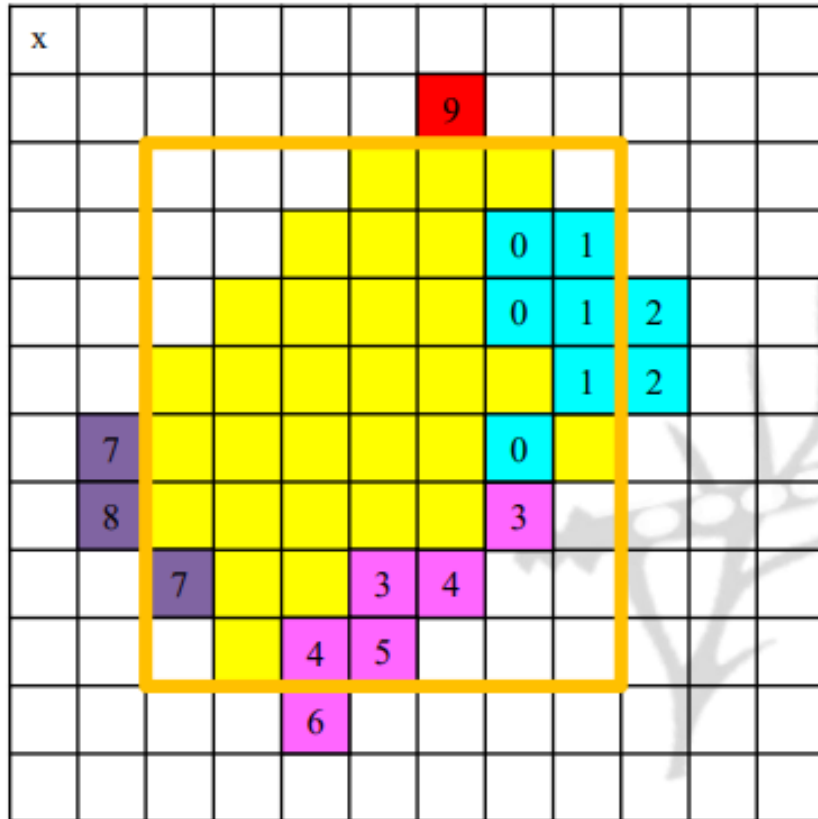


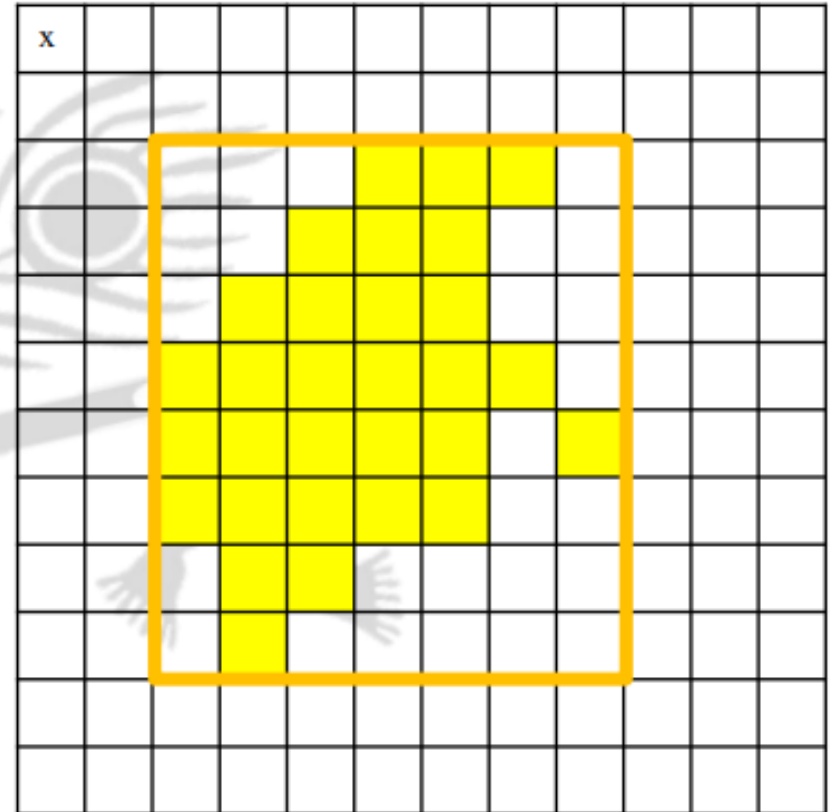
x	x	x
x	0	x
1	1	1

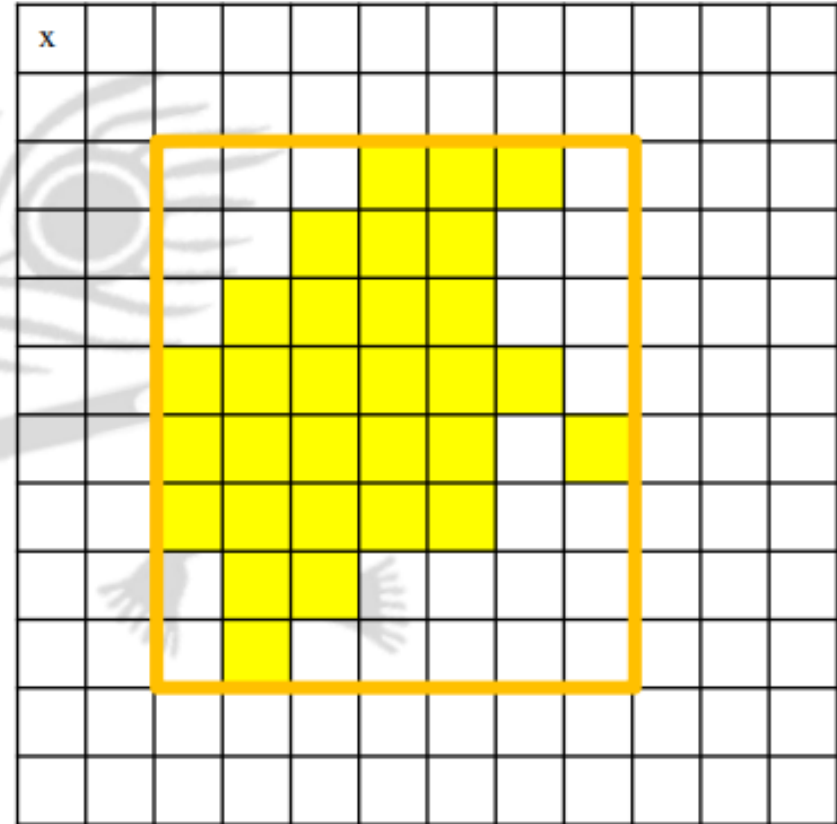
S_3



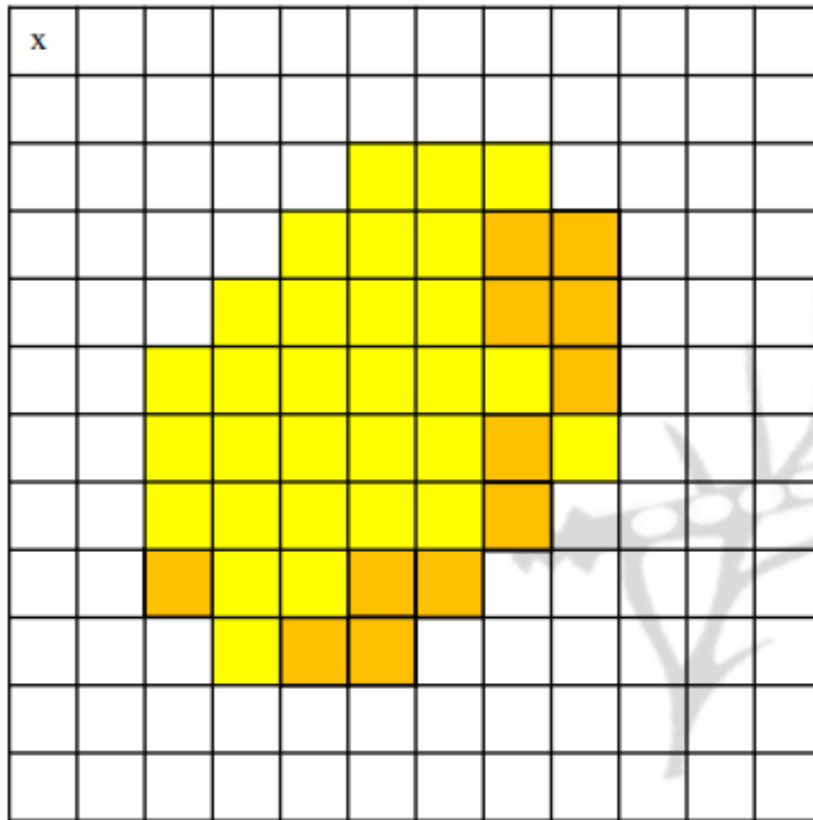
Convex Hull



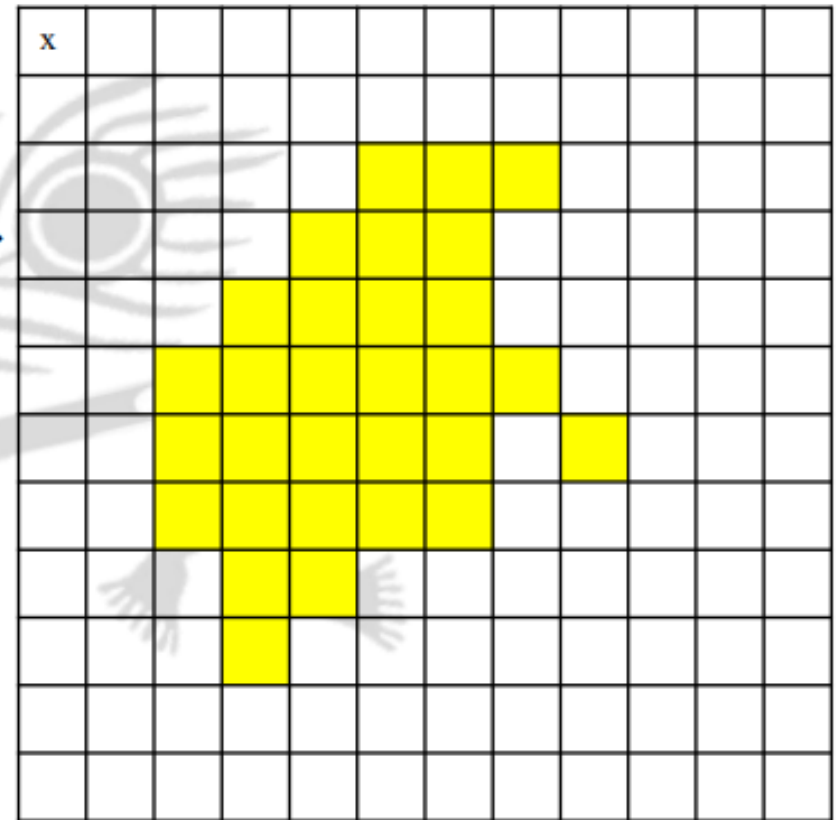




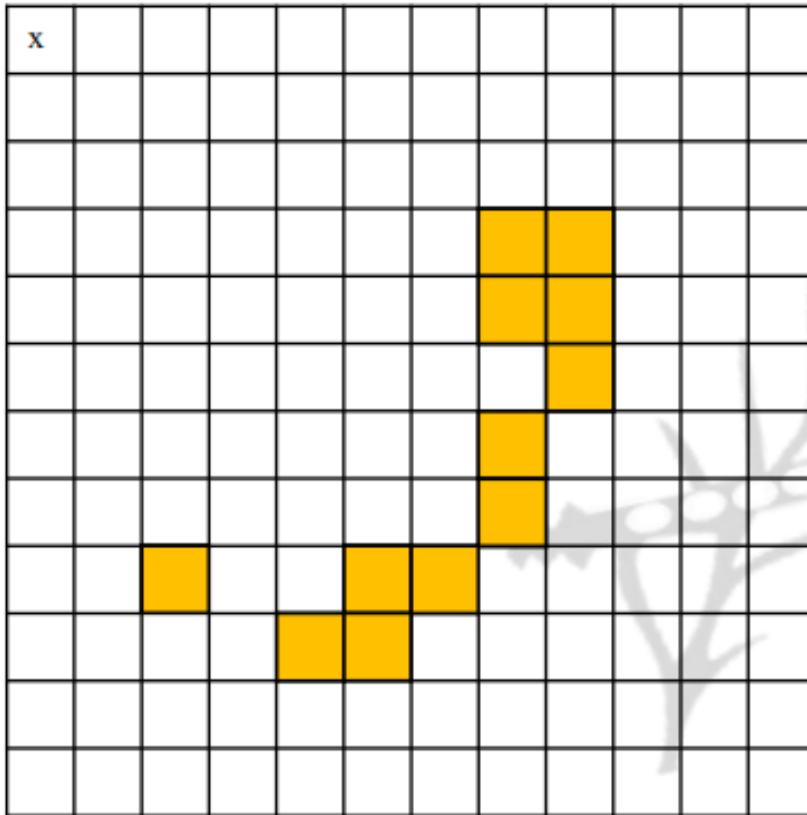
Convex Hull



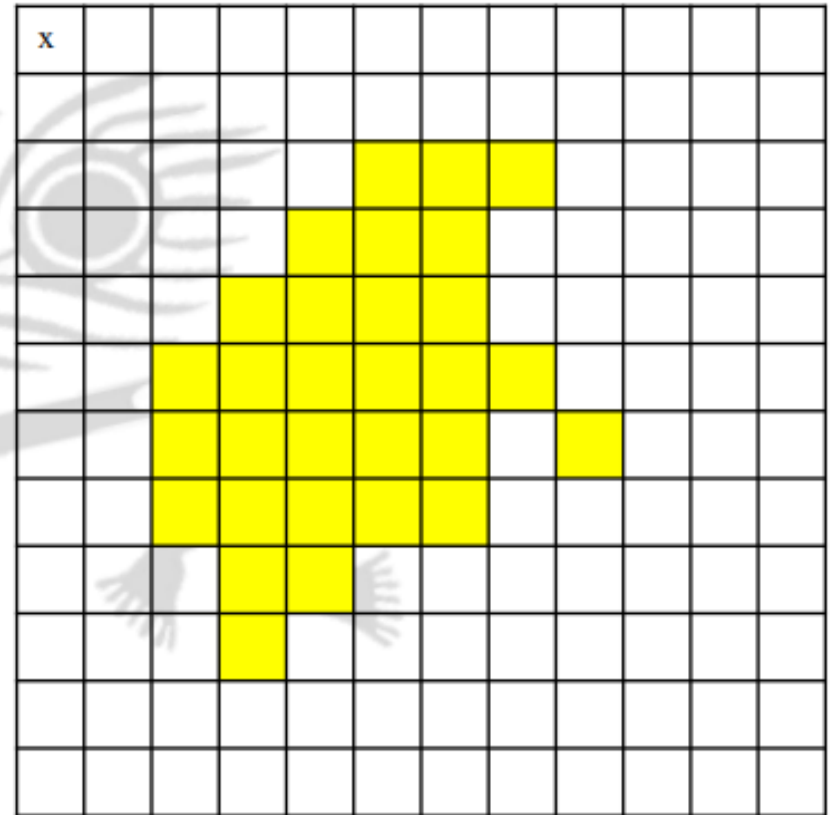
← H
 I →



Convex Hull



←D



Convex Hull

- ❑ Convex Hull (H) = Minimum convex set containing set I
- ❑ Hull Deficiency (D) = $H - I$
- ❑ Steps include –
 - Choose a seed point X_0
 - do $i = 0$ to 3
 - Iterate until convergence
 - Minimize convex set using bounding box of I

University Question

Ques 1: When do you say the shape is Convex? What do you mean by Convex Hull? Give the basic 3×3 Structuring Elements used to find the Convex hull?

GLA University Mathura, End Term Examination, Even Semester, 2022-23

University Question

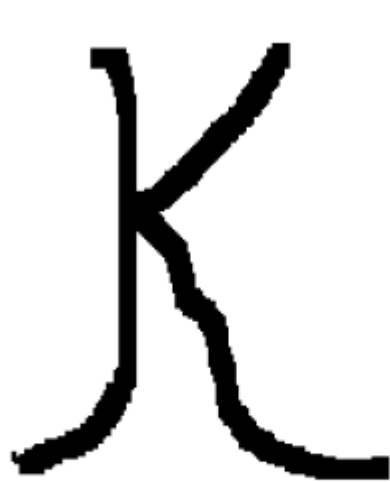
Ques 2: Explain Convex hull with the help of an example.

GLA University Mathura, End Term Examination, Even Semester, 2021-22

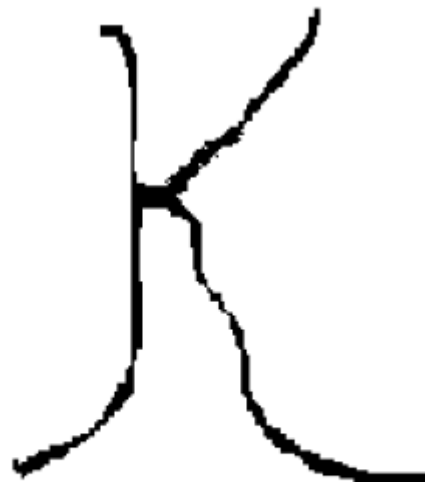
Thinning Operation

Thinning Operation

- ❑ It is an operation that is used to remove selected foreground pixels from binary images
- ❑ It is the process of reducing an object in a digital image to the minimum size necessary for machine recognition of that object
- ❑ It is essentially a “preprocessing step” used in many image analysis techniques



Original image



Thinned image

Thinning Operation

- The thinning of a set A by a structuring element B can be defined in terms of the hit-or-miss transform

$$A \otimes B = A - (A \odot B)$$

- A more useful expression for thinning A symmetrically
- Use a sequence of structuring elements

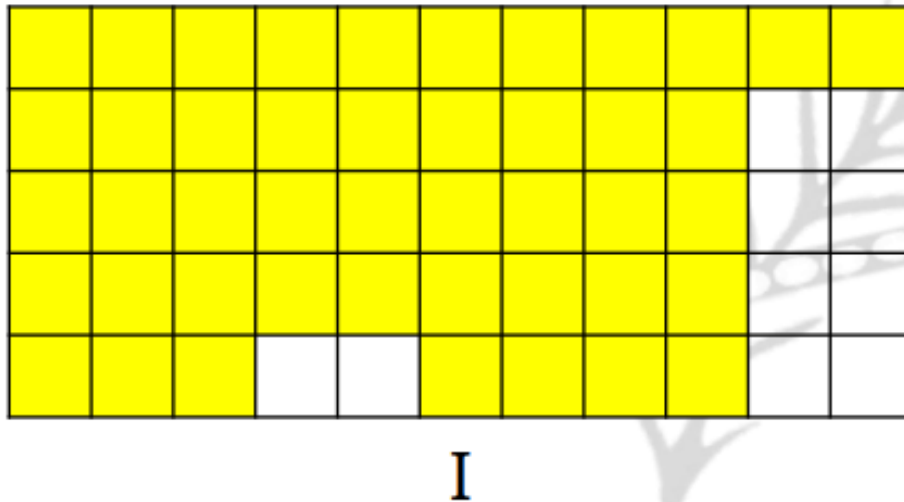
$$\{B\} = \{B^1, B^2, \dots, B^n\}$$

where B^i is the rotated version of B^{i-1}

$$A \otimes \{B\} = \left(\left(\dots \left((A \otimes B^1) \otimes B^2 \right) \dots \right) \otimes B^n \right)$$

Thinning Operation

Example 1:



0	0	0
X	1	X
1	1	1

S_0

X	0	0
1	1	0
1	1	X

S_1

1	X	0
1	1	0
1	X	0

S_2

1	1	X
1	1	0
X	0	0

S_3

1	1	1
X	1	X
0	0	0

S_4

X	1	1
0	1	1
0	0	X

S_5

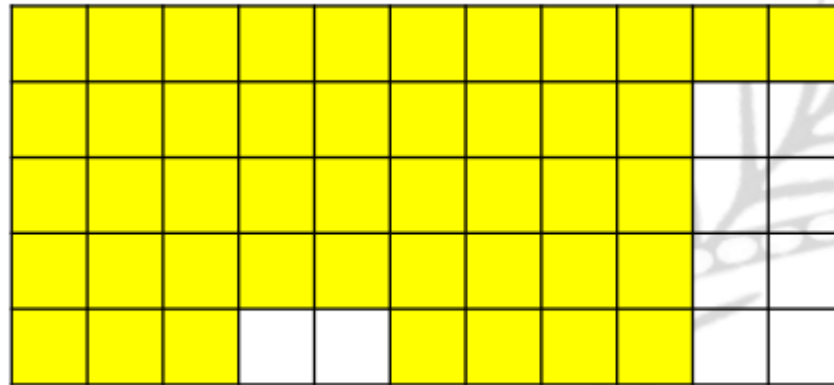
0	X	1
0	1	1
0	X	1

S_6

0	0	X
0	1	1
X	1	1

S_7

Thinning Operation



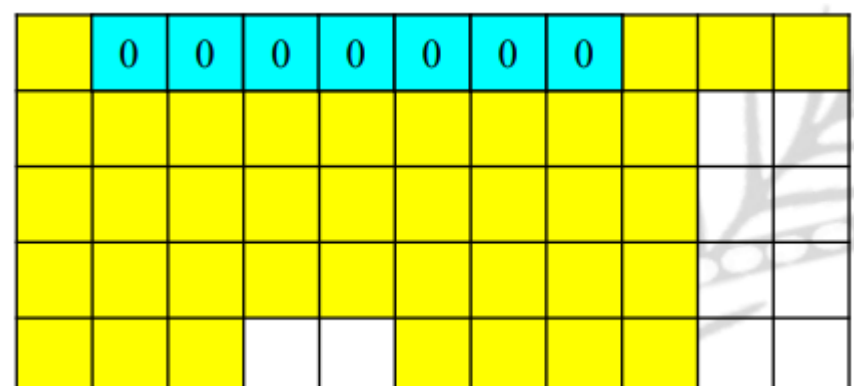
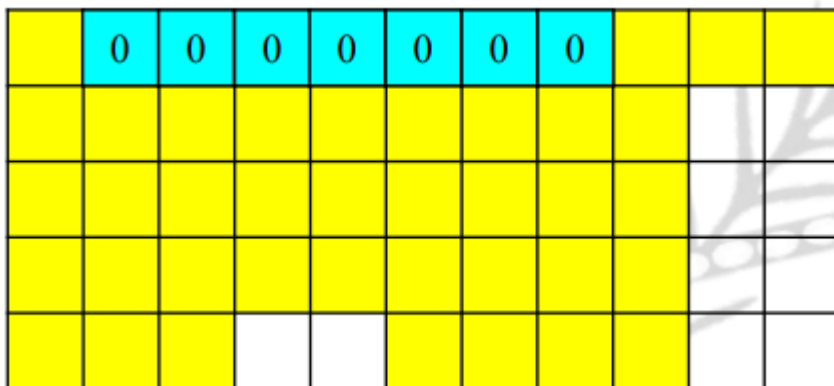
I

0	0	0
X	1	X
1	1	1

S_0

X	0	0
1	1	0
1	1	X

S_1



Thinning Operation

1	X	0
1	1	0
1	X	0

S_2

1	1	X
1	1	0
X	0	0

S_3

	0	0	0	0	0	0	0			
								0		
								0		

	0	0	0	0	0	0	0			
								0		
								0		
		0								

Thinning Operation

1	1	1
X	1	X
0	0	0

S_4

	0	0	0	0	0	0	0			
								0		
			0					0		
	0	0			0	0				

X	1	1
0	1	1
0	0	X

S_5

	0	0	0	0	0	0	0			
								0		
			0	0				0		
	0	0			0	0				

0	X	1
0	1	1
0	X	1

0	0	X
0	1	1
X	1	1

	0	0	0	0	0	0	0			
0								0		
			0	0				0		
	0	0			0	0				

	0	0	0	0	0	0	0			
0								0		
			0	0				0		
	0	0			0	0				

Yellow	White	White	White	White	White	White	White	Yellow	Yellow	Yellow
Yellow	Yellow	Yellow	Yellow	Yellow	Yellow	Yellow	Yellow	Yellow	White	White
White	Yellow	Yellow	Yellow	Yellow	Yellow	Yellow	Yellow	White	White	White
Yellow	Yellow	Yellow	White	White	Yellow	Yellow	Yellow	White	White	White
Yellow	White	White	White	White	White	White	Yellow	Yellow	White	White

0	0	0
X	1	X
1	1	1

So

[illegible]

S_1

Thinning Operation

		0	0	0	0	0				

1	X	0
1	1	0
1	X	0

S_2

Thinning Operation

		0	0	0	0	0				

1	1	X
1	1	0
X	0	0

S_3

Thinning Operation

		0	0	0	0	0				
		0			0					

1	1	1
X	1	X
0	0	0

S_4

Thinning Operation

		0	0	0	0	0				
		0			0	0				

X	1	1
0	1	1
0	0	X

S_5

Thinning Operation

		0	0	0	0	0				
		0			0	0				

0	X	1
0	1	1
0	X	1

S_6

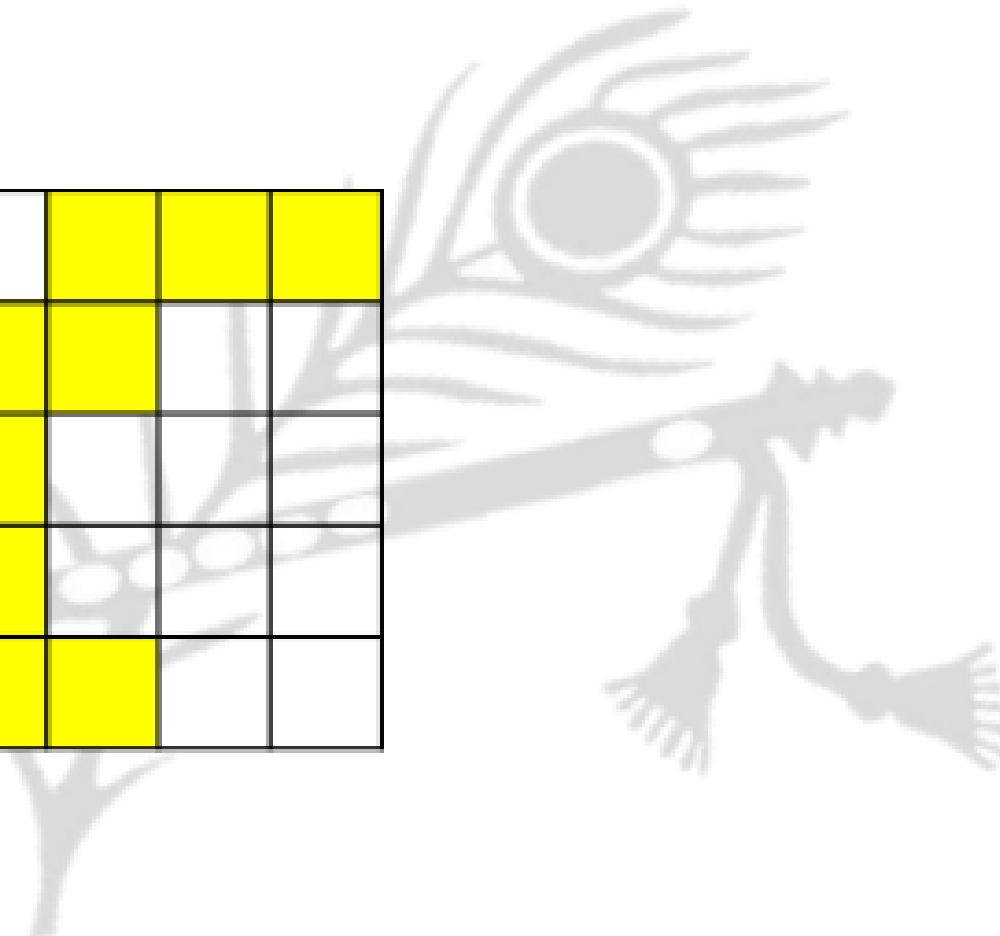
Thinning Operation

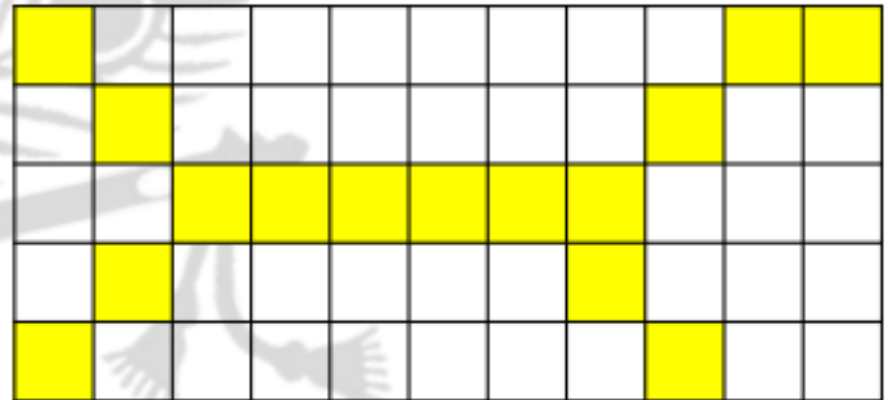
		0	0	0	0	0				
		0			0	0				

0	0	X
0	1	1
X	1	1

S_7

Thinning Operation





Thickening Operation

Thickening Operation

- Thickening is the morphological dual of thinning

$$A \odot B = A \cup (A \circledast B)$$

- Thickening can also be defined as a sequential operation

$$A \odot \{B\} = \left(\left(\dots \left((A \odot B^1) \odot B^2 \right) \dots \right) \odot B^n \right)$$

Thickening Operation

- A separate algorithm is rarely used for thickening
- **The usual procedure??**
- **Thin the background of the set in the question & then take the complement of the result**

To thicken A

- Let $C = A^c$
- Thin C
- Take C^c



Thank You